

Portfolio — System Design

Smart Receipt Automation Platform

Designed and implemented a hybrid event-driven receipt processing system integrating cloud APIs, local automation workflows, OCR processing, and multi-channel ingestion.

The system supports both remote and local usage scenarios:

- Mobile users can upload receipts through the LINE Messaging API.
- Office users can batch-process receipt images by dropping files into monitored local folders.
- Web/API integrations can reuse the same centralized processing pipeline.

Architecture Overview

Multi-Channel Input

- └─ LINE Webhook
- └─ Local Folder Trigger
- └─ Web Upload / API



Workflow Orchestration

- └─ n8n (local automation)
- └─ schedule jobs
- └─ event-driven triggers
- └─ duplicate routing
- └─ notification handling



Centralized Processing API

- └─ Flask REST API
- └─ OCR processing
- └─ receipt parsing
- └─ duplicate detection
- └─ validation logic



Data Layer

- └─ SQLite
- └─ structured receipt records
- └─ notification logs
- └─ reporting source of truth



Reporting / Notification
└─ Google Sheets
└─ duplicate alerts
└─ parse-failure tracking
└─ weekly reports

Key Design Concepts

Hybrid Cloud + Local Architecture

The system intentionally separates:

- cloud processing services
- local workflow orchestration
- user-facing ingestion channels

This design reduces deployment complexity for small businesses while still supporting automation and remote access.

Unified Processing Pipeline

All ingestion methods ultimately reuse the same processing API:

OCR
→ receipt parsing
→ duplicate detection
→ persistence
→ notification routing

This avoids duplicated business logic and simplifies long-term maintenance.

Event-Driven Workflow Design

The system uses event-based automation for:

- file creation triggers
- webhook events
- scheduled reporting
- conditional notification routing

n8n is used as a lightweight orchestration layer, while Flask handles centralized business logic and data processing.

Separation of Concerns

Component	Responsibility
LINE / Folder Trigger	Event ingestion
n8n	Workflow orchestration
Flask API	Business logic
SQLite	Persistent storage
Google Sheets	Notification & reporting

Technical Stack

- Python
- Flask REST API
- SQLite
- n8n
- Docker
- LINE Messaging API
- Google OAuth
- Google Sheets API
- OCR Processing
- Event-Driven Automation

Features Implemented

- Receipt OCR extraction
- Structured receipt parsing
- Duplicate invoice detection
- LINE image upload integration
- Local folder-triggered automation
- Google Sheets notification pipeline
- Parse-failure monitoring
- Weekly reporting API
- REST-based service architecture
- Event-driven workflow orchestration

Engineering Focus

Special attention was placed on:

- maintainability
- encapsulation
- reusable service design
- workflow decoupling
- low-friction SME deployment
- centralized data consistency

The architecture was designed to support future expansion into:

- AI expense classification
- analytics dashboards
- cloud orchestration
- multi-user support
- accounting integrations